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Professor Annalisa Crannell
Editor, *The American Mathematical Monthly*
Mathematical Association of America

Dear Professor Crannell,

I submit for your consideration the manuscript:

The Continuous-Coefficient Jensen Equation:
How Real Mixing Weights Retire Regularity

for publication in *The American Mathematical Monthly* as an **Article**.

What the paper does. The manuscript shows that the continuous-coefficient Jensen equation $G(pu_1 + (1-p)u_2) = pG(u_1) + (1-p)G(u_2)$ for all $u_1, u_2 \in [0, M]$ and all $p \in [0, 1]$ forces G to be affine with *no regularity hypothesis*: a one-line endpoint substitution ($u_1{:=}M$, $u_2{:=}0$, $p{:=}v/M$) reads $G(v)$ directly off the straight line through the endpoints. The manuscript pairs this one-line observation with (i) a dictionary comparing the result with the well-known case of the discrete-coefficient Jensen equation (where Hamel's pathology is real and regularity *is* needed), (ii) a structural explanation of why the two cases differ, and (iii) a diagnostic that separates the saturated-Jensen settings in which these regularity hypotheses are vestigial from the more common settings in which they remain load-bearing. The same five hypotheses appear on both sides of the line: in the axiomatic characterization of Shannon entropy they are indispensable; in a calibration computation on an atomless probability space they are not. A worked quantum example — mixture-affine assignments on density operators — shows the asymmetry at work in a single modern literature: Gleason-type derivations of the Born rule spend positivity or regularity assumptions of the same Cauchy–Hamel kind on the measurement side, while the preparation side, by the convex-domain form of the theorem, spends none.

Why the Monthly. The core result is elementary enough for a second-year analysis student to verify in an afternoon, yet modern applied work spans both sides of a sharp line: in some saturated-Jensen settings the continuous-coefficient equation makes the classical regularity hypotheses vestigial, while in others — superficially identical — those hypotheses remain indispensable. The article supplies the diagnostic that tells them apart, and its closing note on provenance recounts how the author was led to it by an instance of the rarer error of adding a hypothesis that was not needed. This is a natural fit for the *Monthly*: an expository article that makes this asymmetry explicit, mirrors the regularity dictionary, and invites readers to audit their own derivations.

Novelty and relationship to prior art. The substitution technique that closes Theorem 1 is folklore: Aczél's 1966 monograph (§2.1.4) contains a dyadic precursor proved under a continuity hypothesis. The manuscript cites Kuczma only for the classical Cauchy–Jensen regularity catalog, the rational-coefficient pathology, and convex-domain background, not as an explicit source for the exact continuous-coefficient endpoint-substitution statement isolated here. The precise continuous-coefficient regularity-free statement of Theorem 1 does not, however, appear in this exact form in the standard references the author consulted. The contribution of this article is therefore not Theorem 1 itself but its explicit framing: the dictionary contrasting the two Jensen forms, the structural explanation of the Hamel-pathology mechanism, and the identification of a recurrence pattern in derivations that produce the continuous-coefficient form as a saturated identity. In that sense, the article contrasts fully developed classical Cauchy–Jensen routines with a continuous-coefficient use that often remains implicit in applied arguments and is made explicit here. The closing note records that the observation reached the author through a formalization project on a separate topic — a route that may interest readers curious about how mechanized proof can pressure-test classical heuristics.

Submission details.

- **Manuscript type:** Article (expository).
- **Page count:** 19 pages anonymous body; 20 pages with the title page (Monthly LaTeX template formatting).
- **Word count (approximate):** 7,400 words of main text, excluding figures, tables, and references.
- **Figures and tables:** three monochrome figures (drawn in Ti*k*Z, embedded in the LaTeX source per the *Monthly*'s figure instructions) and two tables.
- **References:** 54 entries, typeset in the *Monthly*'s house style (alphabetical, bracketed numbers, DOIs included where available).
- **MSC 2020 primary:** 39B22 (Functional equations for real functions).
- **MSC 2020 secondary:** 26B25 (Convexity), 91B16 (Utility theory), 94A17 (Measures of information; entropy).
- **Double-anonymous:** Yes. The anonymous body (`manuscript-anon.pdf`) contains no author-identifying information; the file with author details (`manuscript.pdf`) carries the title page on page 1.
- **Preprint status:** The manuscript has not been deposited on arXiv. It is not under consideration elsewhere.
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- **Use of generative AI:** Disclosed in the manuscript's Acknowledgments section (AI-assisted exploratory theorem-discovery within a Lean 4 formalization harness, and AI-assisted proof-reading and formatting). All results and their presentation were analyzed and carried out by the author, who assumes full responsibility for correctness, originality, and content.

I look forward to the referees' comments.

Sincerely,

Ali Elouafiq